

Safety and Reliability Training

Module 1: General Introduction

Submodule 1.B

INTRODUCTION TO AIRCRAFT CERTIFICATION

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ANZEN

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Objectives of this training

- ✓ Get to know the main Aircraft Certification concepts
 - ✓ To understand the general Airworthiness environment
 - ✓ To achieve the capability to go through regulations and specifications
 - ✓ To understand the Certification Process
- ❖ This Submodule presents the case of manned vehicles. Particularities for unmanned will be covered in an Additional Training Module.



Outline

- ☐ Introduction
 - ☐ The need for Certification
 - ☐ Preliminary concepts
- ☐ Regulatory Framework
- ☐ Part 21
- ☐ Airworthiness Controlled Environment
- ☐ Airworthiness Codes: CS-25
- ☐ Certification Process



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- ☐ **Introduction**
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Introduction: The need for Certification (I)

- **Regulation (EU) 2018/1139:**

*Article 11: The **design of a product** shall be **subject to certification** and shall be issued with a **type certificate** [...]. Article 14 (1): **Individual aircraft** shall be **subject to certification** and shall be issued with a **certificate of airworthiness** [...].*

- **Ley 48/1960 sobre Navegación Aérea:**

*Artículo 36: Ninguna aeronave, salvo las exceptuadas en el artículo 151 de esta Ley, será autorizada para el vuelo sin la previa expedición de un **certificado de aeronavegabilidad** [...]. Compete al Ministerio de Fomento extender el certificado de aeronavegabilidad de las aeronaves civiles [...].*

- **Reglamento de Aeronavegabilidad de la Defensa:**

Preámbulo: [...] debe entenderse, que dicha competencia la tiene el Ministerio de Defensa para las aeronaves que tengan la consideración de militares o que pertenezcan a los Ejércitos, institutos, organismos o servicios dependientes o vinculados al Ministerio de Defensa.

- **Convention on International Civil Aviation (Chicago, 1944):**

*Article 31: Every aircraft engaged in international navigation shall be provided with a **certificate of airworthiness** issued or rendered valid by the State in which it is registered.*

Introduction: The need for Certification (II)

- In aviation, pertinent Certificates, issued by the Competent Authority, are required by law and regulations:

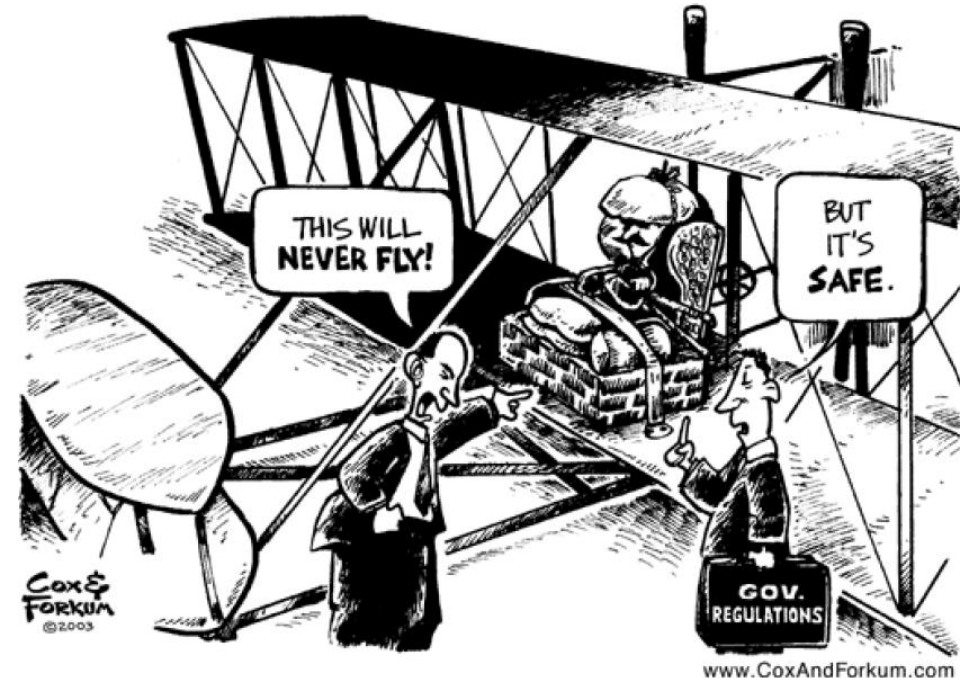
Certificate of Airworthiness for Individual aircraft.

Type Certificate for the design of a product.

- A **Certification Process** must be developed to demonstrate the “fitness to flight” (or condition for safe operation).

❖ Safety first

❖ Bearing in mind the final purpose



Introduction: Preliminary concepts (I)

- Airworthiness (AW)

- ❖ Airworthiness may be defined as the fitness of an aircraft for flight in all conditions (environmental, operational,...) for which it has been designed and to which it may therefore be exposed. An airworthy airplane is one which is **fit to fly**.
- ❖ Alternative definition: An airworthy aircraft is one where the likelihood of any incident or accident as a result of malfunction, performance or handling of the aircraft is kept to acceptable levels (defined in the authorities' requirements).

The only real measure of airworthiness in use is given by tracking and analyzing incidents and accidents.

Introduction: Preliminary concepts (II)

- **Initial AW**

- ❖ The Certification Process developed to demonstrate the “fitness to flight” of a design is also known as Initial Airworthiness.

- **Continued AW**

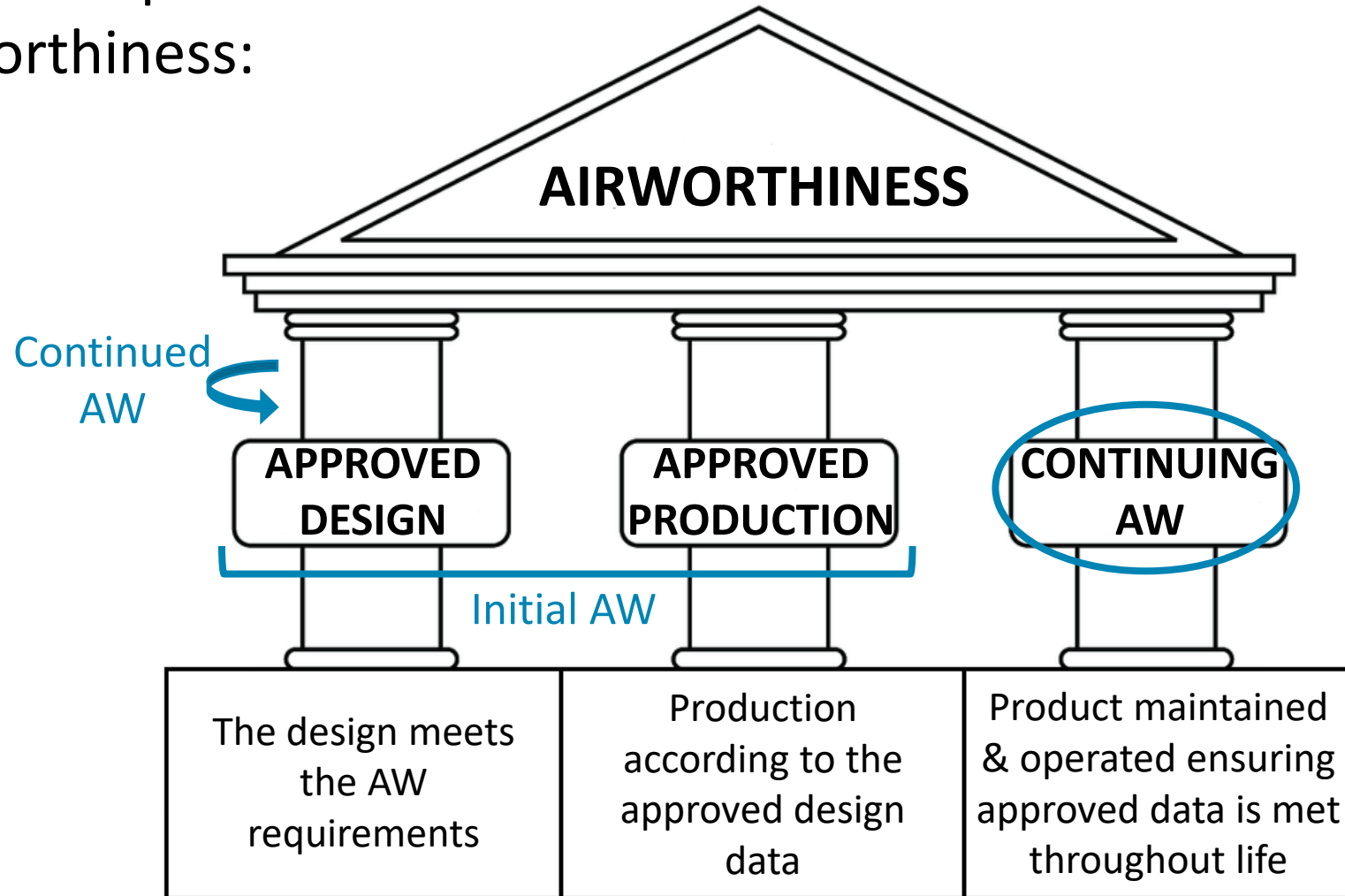
- ❖ All the actions associated with the upkeep of a type design and the associated approved data through life. (*“Aeronavegabilidad Continuada”*)
- ❖ Responsibility of the manufacturer (Competent Authority: State of Design)

- **Continuing AW**

- ❖ All the processes ensuring that, at any time in its operating life, the aircraft complies with the airworthiness requirements in force and is in a condition for safe operation. (*“Mantenimiento de la Aeronavegabilidad”*)
- ❖ Responsibility of the operator (Competent Authority: State of Registry)

Introduction: Preliminary concepts (III)

- The three pillars of Airworthiness:



Introduction: Preliminary concepts (IV)

- **Certification Rules structure:**

- ❖ **Requirements** -> When applicable, mandatory.

- *What is required? What should be demonstrated?*

- ❖ **Advisory Material** -> Associated to requirements.

- *How the requirement should be interpreted? How can it be demonstrated? Which is the scope of the requirement? ...*

- Interpretative material (IM)
 - Acceptable means of compliance (AMC)
 - Guidance material (GM)
 - Advisory Circular (AC)



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Regulatory Framework: ICAO (I)

ICAO (OACI)

- *The International Civil Aviation Organization (ICAO) is a United Nations agency which **helps** 193 countries to cooperate together and share their skies to their mutual benefit.*
- General objective: **Support and coordinate countries** (diplomatically and technically) to:
 - Set a uniquely rapid and dependable network of global air mobility
 - Promoting sustainable growth and socio-economic prosperity wherever aircraft fly
- Strategic Objectives:
 - ✓ Safety
 - ✓ Air Navigation Capacity and Efficiency
 - ✓ Security & Facilitation
 - ✓ Economic Development of Air Transport
 - ✓ Environmental Protection



([Home \(icao.int\)](http://icao.int))

Regulatory Framework: ICAO (II)

- **Convention on International Civil Aviation**

*“THEREFORE, the undersigned governments having agreed on certain **principles and arrangements** in order that **international civil aviation** may be developed in a safe and orderly manner and that international air transport services may be established on the basis of equality of opportunity and operated soundly and economically” (Chicago, 1944)*

96 articles structured in 4 parts

PART I: AIR NAVIGATION

PART II: THE INTERNATIONAL CIVIL AVIATION ORGANIZATION

PART III: INTERNATIONAL AIR TRANSPORT

PART IV: FINAL PROVISIONS

+ 19 annexes [ICAO Annexes and Doc Series](#) | [SKYbrary Aviation Safety](#)

SARP (Standards and Recommended Practices)

- Standards: the uniform application is recognized as necessary
- Recommended: the uniform application is recognized as desirable

PANS (Procedures for Air Navigation Services)

SUPP (Regional Supplementary Procedures)

Guidance material

Regulatory Framework: ICAO (III)

In summary:

- ICAO (OACI) is not an Aviation Authority
- It provides a global forum of States for international civil aviation:
 - Provides support and coordination
 - Promotes cooperation
 - Develops policies and standards, conducts compliance audits, performs studies and analyses,...
- Member States have obligations and responsibilities:
 - Compliance with the Convention of primary aeronautical legislation
 - Specific operating regulations under SARPS
 - Development and promulgation of civil aviation regulations
 - Establishment of a Civil Aviation Authority

Regulatory Framework: EASA (I)

EASA (European Aviation Safety Agency)



- **Background:**
 - Civil Aviation National Authorities were responsible of the implementation and control of ICAO SARPs in their respective national regulations.
 - JAA (Joint Aviation Authorities):
 - Not an Authority itself, but a "club of authorities" (EU and non-EU).
 - Development of common rules: **JAR** (Joint Aviation Regulations).
 - Not within UE regulatory framework.
- **EASA creation** with the following **mission**:
 - Ensure the highest common level of safety protection for EU citizens.
 - Ensure the highest common level of environmental protection.
 - Single regulatory and certification process among Member States.
 - Facilitate the internal aviation single market & create a level playing field.
 - Work with other international aviation organizations & regulators.

Regulatory Framework: EASA (II)

- **Distribution of competences:**

EASA:

- Type Certificates and approval of design modifications
- Design Organizations approval

National Authorities:

- Production/Maintenance/Maintenance Training Organizations approval
- Issue of licenses

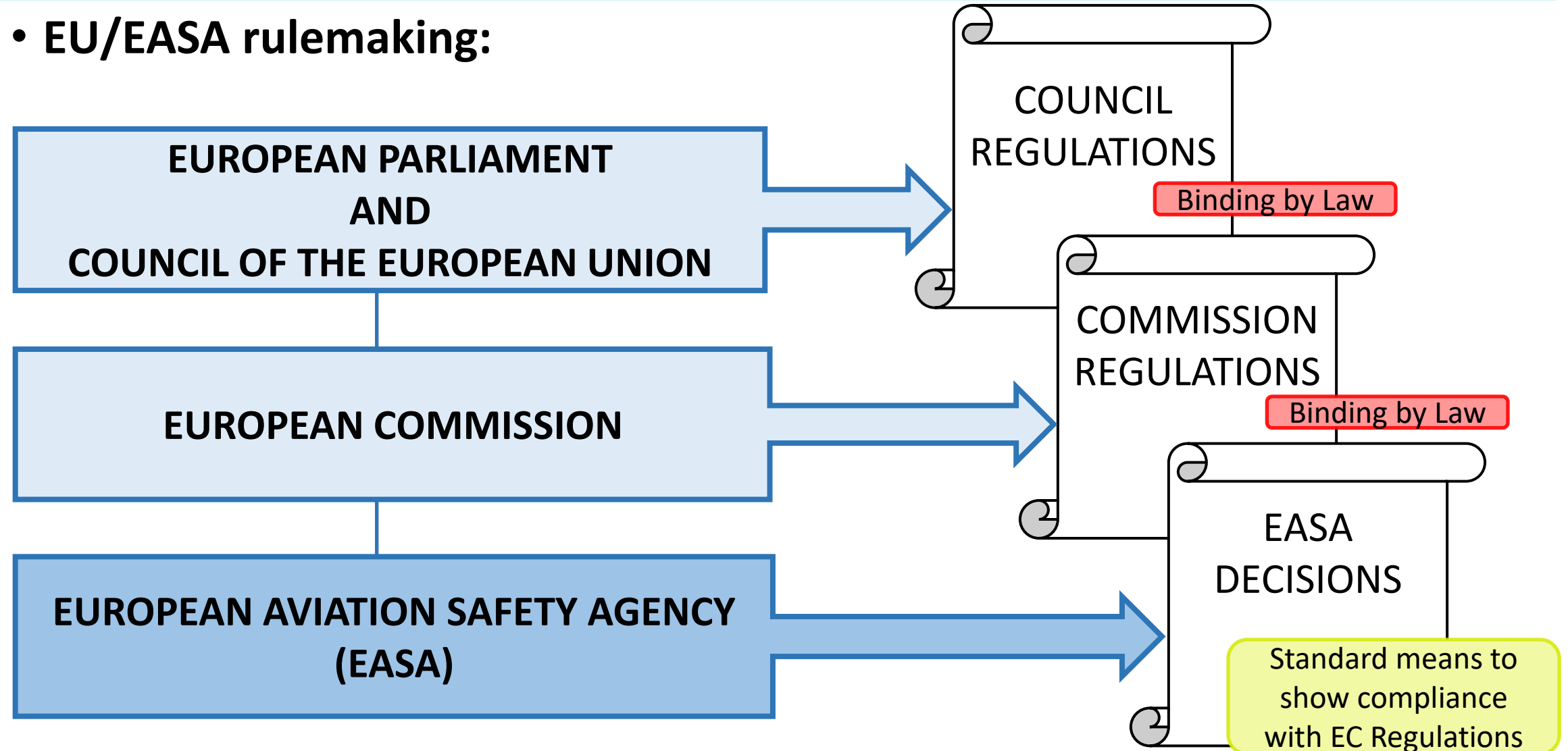
➤ **Spain:**

AESA (*Agencia Estatal de Seguridad Aérea*): Ensures compliance of air transport safety standards in Spain (Supervision, Inspection, Management, Sanctioning...)

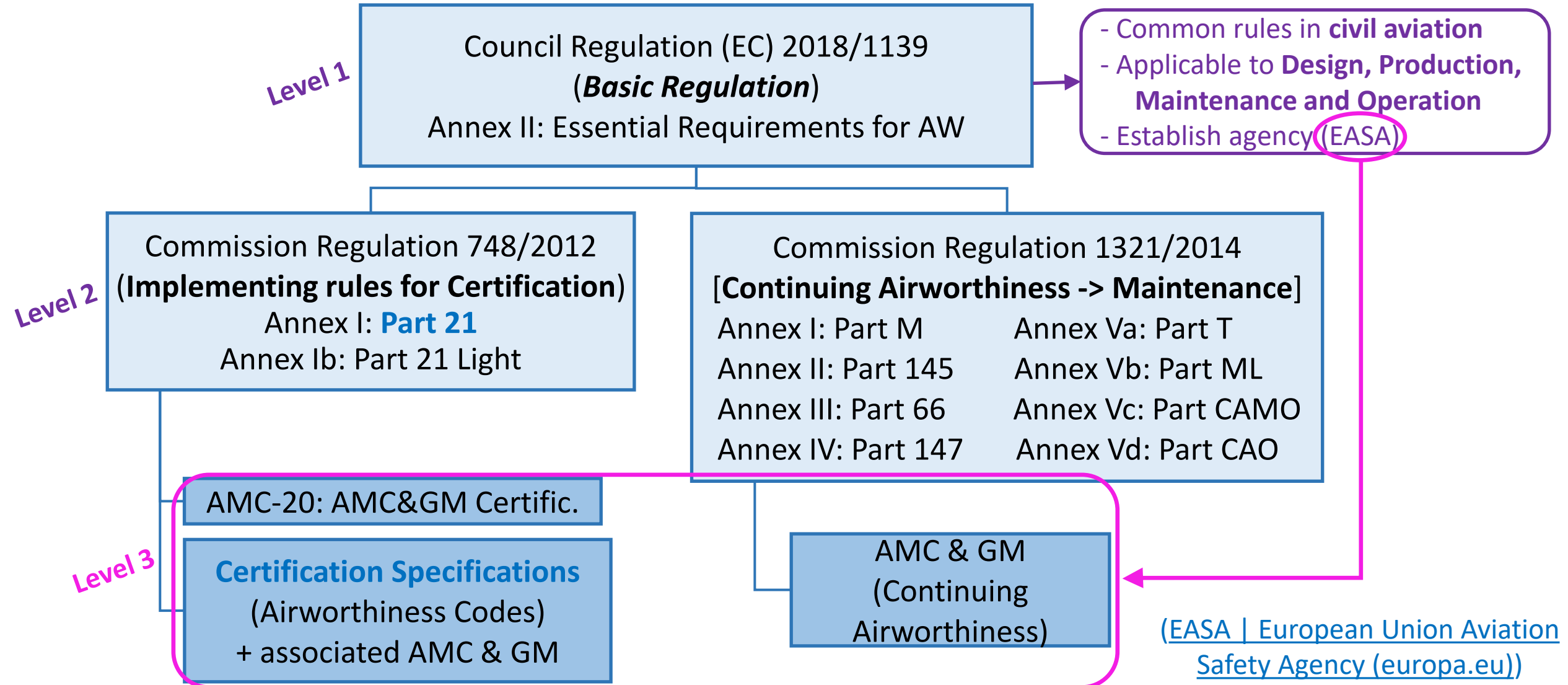
DGAC (*Dirección General de Aviación Civil*): Acts as a regulator within the national competences (designing strategies and policies, coordination with other administrations, etc.)

Regulatory Framework: EASA (III)

- EU/EASA rulemaking:



Regulatory Framework: EASA (IV)



Regulatory Framework: EDA (I)

EDA (European Defence Agency) (<https://eda.europa.eu/home>)



- Defence is and will remain a national responsibility, but the EU can help its 27 members to buy, develop and operate new assets together (save money, work together, reinforce NATO).
- EDA was created in 2004 to **promote defense collaboration in the EU**.
- Three **main missions**:
 - Support the development of defence capabilities and military cooperation.
 - Stimulate defence Research and Technology and strengthen the European defence industry.
 - Act as a military interface to EU policies.
- Organization:
 - Under the authority of the Council of the EU, to which it reports and from which it receives guidelines.
 - Defence Ministers (Steering Board) decide work plan, projects, programs and new initiatives.
 - Working groups comprised of national experts can be formed for any given subject.
 - Specific thematic fora -> **EDA's Military Airworthiness Authority (MAWA) Forum**.

Regulatory Framework: EDA (II)

- The **MAWA Forum**, created in 2008, strives to **harmonize the airworthiness requirements and processes** of the EDA pMS (participant Member States).
- It consists of representatives from each of the EDA pMS **NMAAs** and is chaired by the EDA, which also provides the organizational and administrative support.
- The goals of the MAWA Forum are to develop and achieve the following ministerial approved 'roadmap-objectives' for adoption into pMS's regulatory systems:
 - common regulatory framework
 - common certification processes
 - common certification/design codes
 - formation of a European Military Joint Airworthiness Authorities Organisation (EMJAAO)
 - common approach to organisational approvals
 - common approach to preservation of airworthiness
 - arrangements for mutual recognition
- In order to achieve these objectives, the MAWA Forum has created links with external bodies and organizations (including EASA, NATO and industry) to ensure a coherent approach to their activities and to avoid unnecessary duplication.

Regulatory Framework: EDA (III)

- The MAWA Forum is responsible of the development of the **European Military Airworthiness Requirements (EMAR)**: (<https://eda.europa.eu/experts/airworthiness>)
 - ❑ European Military Airworthiness Basic Framework Document (BFD)
 - Annex B: Essential Requirements for AW of manned aircraft
 - Annex C: Essential Requirements for AW of unmanned aircraft
 - ❑ **EMAR 21** (+ AMC & GM)
 - ❑ EMAR 145 (+ AMC & GM)
 - ❑ EMAR 66 (+ AMC & GM)
 - ❑ EMAR M (+ AMC & GM)
 - ❑ EMAR CAMO (+ AMC & GM)
 - ❑ EMAR 147 (+ AMC & GM)
 - ❑ European Military Airworthiness Certification Criteria (**EMACC**)
 - ❑ EMAR Implementation Guides
 - ❑ EMAR Forms Document
 - ❑ EMAD 20
 - ❑ EMAD 1
 - ❑ EMAD MFTP
 - ❑ EMAD R

Regulatory Framework: Other Authorities (I)

- Federal Aviation Administration (FAA) -> FAR
- Luftfahrt Bundesamt (LBA)
- Direction Générale de l'Aviation Civile (DGAC)
- Bundesamt für Wehrtechnik und Beschaffung (BWB)
- Direction Générale de l'Armement (DGA)

❖ International Military Programs particularities:

- Industry cooperation
- Several authorities involvement
 - Different regulatory frameworks
 - Need of groups, panels and committees
- Military topics to be covered

Regulatory Framework: Other Authorities (II)

Spanish MoD:

- *RAD: Reglamento de Aeronavegabilidad de la Defensa (RD 866/2015):*
 - ❖ **AAD** (Autoridad de AW de la Defensa) -> **DiGAM** (Director General de Armamento y Material)
 - ❖ Asesorado por el Consejo de Aeronavegabilidad
 - ❖ Órgano Técnico Competente (OTC): Emite “Certificado Técnico”
 - Subdirección General de Experimentación y Certificación del INTA (-> SG SSAA)
 - Subdirección de Inspección, Regulación y Estrategia Industrial de Defensa de la DGAM.
- *Procedimientos (Consejo, ET, EA, ARM, INTA, GC, DGAM/INREID)*
- **Publicaciones Españolas de Requisitos de Aeronavegabilidad Militares (PERAM):**
Instrumento oficial del Ministerio de Defensa para definir requisitos de aeronavegabilidad militares

- | | |
|--|------------------------------------|
| ☐ PERAM 21 | ☐ PERAM 145 |
| ☐ PERAM 66 | ☐ PERAM 147 |
| | ☐ PERAM M |



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Part 21: General Overview (I)

- Initial Airworthiness and Environmental Protection (Regulation (EU) No 748/2012) - ANNEX I: **PART 21**

Certification of aircraft and related products, parts and appliances, and of design and production organisations

- ❑ 21.1. Competent authority
- ❑ 21.2 Scope
- ❑ **SECTION A — TECHNICAL REQUIREMENTS**
SUBPARTS A-Q (21.A.XX)
- ❑ **SECTION B — PROCEDURES FOR COMPETENT AUTHORITIES**
SUBPARTS A-Q (21.B.XX)
- ❑ Appendix I-XII

([EASA | European Union Aviation Safety Agency \(europa.eu\)](https://easa.europa.eu/))

Part 21: General Overview (II)

SUBPART A — GENERAL PROVISIONS

SUBPART B — TYPE-CERTIFICATES AND RESTRICTED TYPE-CERTIFICATES

(SUBPART C — NOT APPLICABLE)

SUBPART D — CHANGES TO TYPE-CERTIFICATES AND RESTRICTED TYPE-CERTIFICATES

SUBPART E — SUPPLEMENTAL TYPE-CERTIFICATES

SUBPART F — PRODUCTION WITHOUT PRODUCTION ORGANISATION APPROVAL

SUBPART G — PRODUCTION ORGANISATION APPROVAL

SUBPART H — CERTIFICATES OF AIRWORTHINESS AND RESTRICTED CERTIFICATES OF AIRWORTHINESS

SUBPART I — NOISE CERTIFICATES

SUBPART J — DESIGN ORGANISATION APPROVAL (DOA)

SUBPART K — PARTS AND APPLIANCES

(SUBPART L — NOT APPLICABLE)

SUBPART M — REPAIRS

(SUBPART N — NOT APPLICABLE)

SUBPART O — EUROPEAN TECHNICAL STANDARD ORDER AUTHORISATIONS

SUBPART P — PERMIT TO FLY

SUBPART Q — IDENTIFICATION OF PRODUCTS, PARTS AND APPLIANCES

Part 21: Subpart A - General Provisions

21.A.1 Scope: *This Subpart establishes the **general rights and obligations of the applicant for, and holder of**, any certificate that has been issued or is to be issued in accordance with this Annex.*

21.A.3A **Reporting system** -> *Occurrences management*

21.A.3B **Airworthiness directives** -> *Unsafe condition*

AMC1 21.A.3B(b) Failures, malfunctions and defects

GM1 21.A.3B(b) Failures, malfunctions and defects

GM 21.A.3B(d)(4) Defect correction – Sufficiency of proposed corrective action

21.A.4 **Coordination between design and production**

21.A.5 **Record-keeping**

21.A.6 **Manuals**

21.A.7 **Instructions for continued airworthiness**

21.A.9 **Access and investigation**

Part 21: Subpart B - TC and RTC

- Scope, Eligibility, Demonstration of capability, Application,...
- Changes requiring a new type-certificate
 - ❖ *Substantial changes* (the change in design, power, thrust, or mass is so extensive that a substantially complete investigation of compliance with the applicable type CCBB is required)
- **Certification Process**
 - ❖ Compliance demonstration item (CDI) → Level of Involvement (LoI)
- Operational suitability data (**OSD**) → part of an aircraft TC (shall be available):
 - Pilot type rating training, including determination of type rating
 - Data to support qualification of simulators
 - Maintenance certifying staff type rating training
 - Type or variant for cabin crew and type-specific data for cabin crew
 - Master minimum equipment list
- Type design, Type Certificate
- Obligations of the holder
- Transferability, duration and continued validity

Part 21: Subparts D and E

SUBPART D: CHANGES TO TCs AND RTCs

- Classification of changes to a type-certificate -> *minor/major changes*
 - ❖ A “minor change” has no appreciable effect on the mass, balance, structural strength, reliability, operational characteristics, operational suitability data, or other characteristics affecting the airworthiness of the product or its environmental characteristics.
 - ❖ All other changes are “major changes” under this Subpart.
- GM 21.A.101 Establishing the certification basis of changed aeronautical products
 - ❖ Significant change:
 - i. the general configuration or the principles of construction are not retained
 - ii. the assumptions used for certification of the product to be changed do not remain valid

SUBPART E: SUPPLEMENTAL TYPE-CERTIFICATES

- ❖ In general, different manufacturer and/or authority.
- ❖ Legacy documentation/information availability?
- ❖ Example: A330 MRTT, A330 VIP, STCs CETA...





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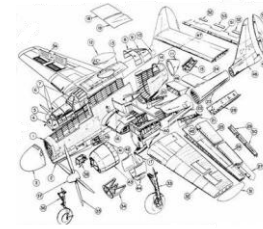
Airworthiness Controlled Environment (I)

DESIGN

DOA
according to
Subpart 21 J

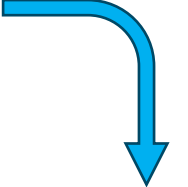


TC issued
according
to Part 21

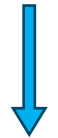


PRODUCTION

POA
according to
Subpart 21G



**AW
Certificate**
for each
aircraft



MAINTENANCE

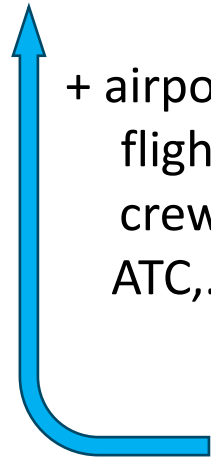


Training organization
according to Part 147,
licenses according to Part 66

Maintenance
organization
according to
Part 145

Operator with
CAMO approved
according to Part M

+ airports,
flight
crew,
ATC,...

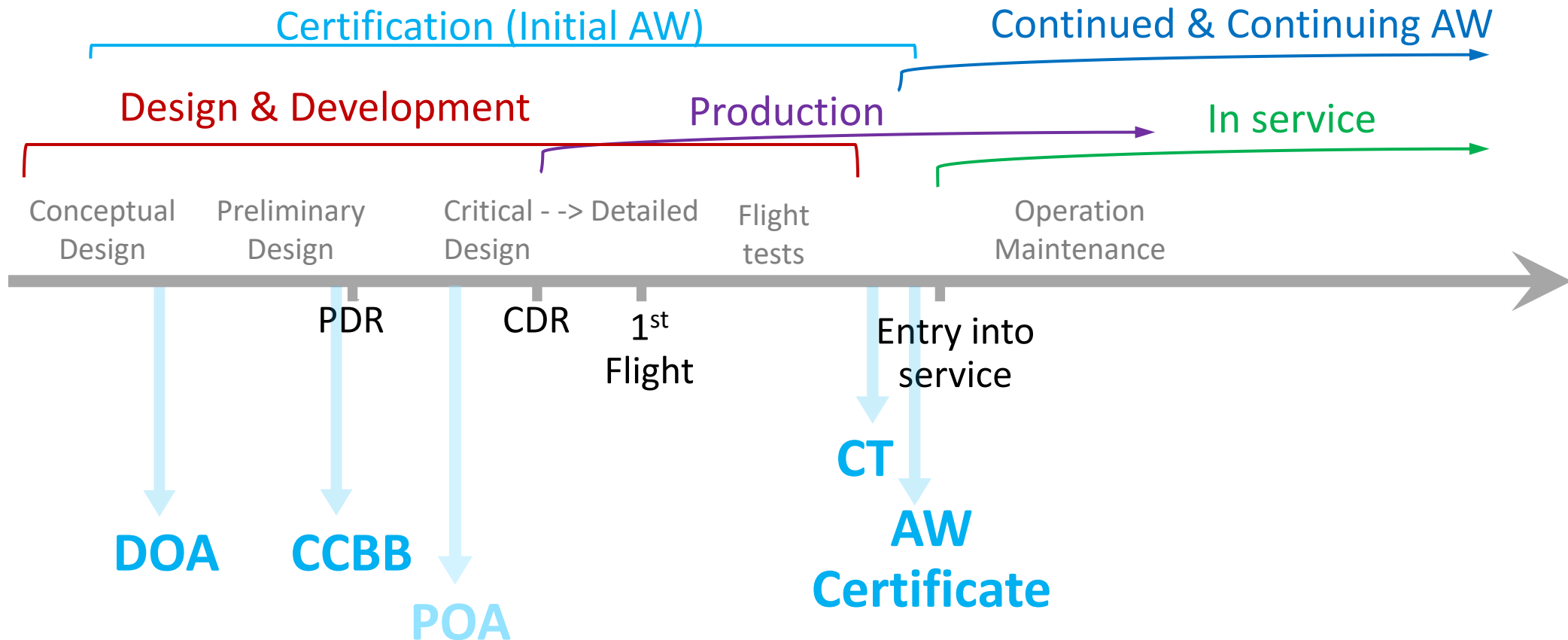


OPERATION



Airworthiness Controlled Environment (II)

LIFE CYCLE





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Airworthiness Codes

- Standards and Regulations from Regulatory Authorities to be used as technical requirements in the Certification Process.
- EASA -> Certification Specifications ([Initial Airworthiness | EASA \(europa.eu\)](#))
 - CS-23: Normal-Category Aeroplanes
 - CS-25: Large Aeroplanes
 - CS-27: Small Rotorcraft
 - CS-29: Large Rotorcraft
 - CS-E: Engines
 - CS-P: Propellers
- JAA -> JAR: Joint Aviation Regulations
- FAA -> FAR: Federal Aviation Regulations ([Federal Aviation Authority](#))

CS-25: CS and AMC for Large Aeroplanes

SUBPART A — GENERAL

SUBPART B — FLIGHT

SUBPART C — STRUCTURE

SUBPART D — DESIGN AND CONSTRUCTION

SUBPART E — POWERPLANT

SUBPART F — EQUIPMENT

SUBPART G — OPERATING LIMITATIONS AND INFORMATION

SUBPART H — ELECTRICAL WIRING INTERCONNECTION SYSTEMS

SUBPART I — NOISE CERTIFICATES

SUBPART J — AUXILIARY POWER UNIT INSTALLATIONS

APPENDIX A-S

GENERAL AMCs

CS-25 amendment 28



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- ❑ **Certification Process**

❖ This section is focused on Aircraft TC. Equivalent process for approval of changes, STCs, or for other products.

Certification Process

- Before a newly developed aircraft model may enter into operation, it must obtain a **Type Certificate** from the responsible aviation regulatory authority.
This certificate testifies that the type of aircraft meets the AW requirements set.
- The 4 main steps of the type-certification process are:
 1. Technical Familiarisation and Certification Basis.
 2. Establishment of the Certification Programme
 3. Compliance demonstration
 4. Technical closure and issue of approval

1. Technical Familiarisation and Certification Basis

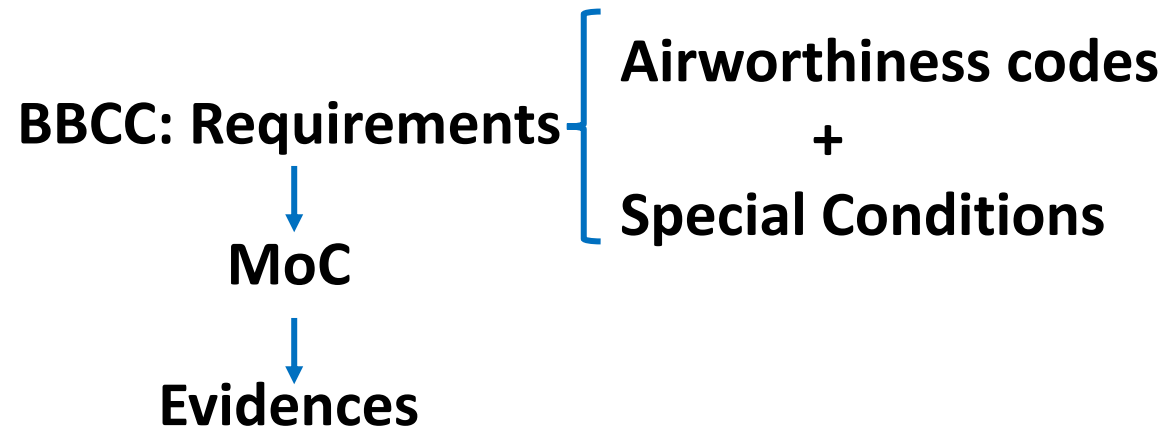
- The aircraft manufacturer presents the project to the authority when it is considered to have reached a sufficient degree of maturity.
- If appropriate, preliminary proposal of Certification Basis.

Certification Process: Certification Programme

2.Establishment of the Certification Programme

(AMC 21.A.15(b) Content of the certification programme)

- Certification Programme Plan
- Description of the product and scope of Certification.
- Agreement of the Certification Basis.
- Definition and agreement of the Means of Compliance (MoC).
- Proposal of the Evidence Documentation.
- If applicable, identification of “level of involvement” (LOI).



Certification Process: Certification Basis

Special conditions (21.B.75): Special detailed technical specifications defined when the related AW code do not appropriately cover the type design:

- the product has novel or unusual design features not covered by the AW code
- the intended use of the product is unconventional
- experience from similar products in service, similar design features or newly hazards show that unsafe conditions may be developed (e.g. Lithium batteries)

Equivalent Level of Safety (21.B.80(a)(2))

- If the applicable requirements cannot be complied with, either fully or partially, it may be accepted a suitable alternative which provides an equivalent level of safety:
 - Using appropriate compensating factors
 - And/or the objectives are met by designs or features other than those required

Certification Review Item (CRI) -> Discussion to reach an agreed SC or ELS.

Certification Process: MoC and Evidences

MoC: Means of Compliance (*Appendix A to AMC 21.A.15(b) Means of compliance codes*)

Type of compliance	Means of compliance	Associated compliance documents
Engineering evaluation	MC0: (a) compliance statement (b) reference to design data (c) election of methods, factors, etc. (d) definitions	(a) Design data (b) Recorded statements
	MC1: design review	(c) Descriptions (d) Drawings
	MC2: calculation/analysis	(e) Substantiation reports
	MC3: safety assessment	(f) Safety analysis
Tests	MC4: laboratory tests	(g) Test programmes (h) Test reports (i) Test interpretations
	MC5: ground tests on related product(s)	
	MC6: flight tests	
	MC8: simulation	
Inspection	MC7: design inspection/audit	(j) Inspection or audit reports
Equipment qualification	MC9: equipment qualification	Note: Equipment qualification is a process that may include all previous means of compliance at equipment level.

Certification Process: Demonstration

3.Compliance Demonstration

- The aircraft manufacturer must demonstrate compliance of its product with the CCBB requirements.
- Issue of compliance evidence documents
(AMC 21.A.20(c) Compliance documentation)
- Compliance matrix
- Authority experts perform a detailed examination of this compliance demonstration by means of document reviews and by attending some of the compliance demonstrations (test witnessing).
- Final declaration of compliance
- This is the longest phase of the type-certification process. In the case of large aircraft, the period to complete the compliance demonstration is set at five years and may be extended, if necessary.

Certification Process: Type Certificate

4. Technical closure and issue of approval

- If technically satisfied with the compliance demonstration by the manufacturer, the Authority closes the investigation and issues the certificate.
- The type-certificate shall include
 - type design (every model and configuration)
 - applicable type-certification basis
 - operating limitations
 - instructions for continued airworthiness
 - environmental protection requirements
 - any other conditions or limitations prescribed for the product
 - operational suitability data
 - type-certificate data sheet for noise
 - record of emissions compliance
- **Type Certificate Data Sheet (TCDS)**

Any questions?



Safety and Reliability
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